

Algebraic Geometry

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0 Introduction/Motivation

Example (Pythagorean Theorem). For a right triangle, $x^2 + y^2 = z^2$. Problem: What are $(x, y, z) \in \mathbb{Z}^3$ where $xyz \neq 0$ such that $x^2 + y^2 = z^2$. Let $X = \frac{x}{z}$, $Y = \frac{y}{z}$. We are looking for rational X, Y such that $X^2 + Y^2 = 1$. Taking a line with rational gradient passing through $(-1, 0)$ say $Y = t(X + 1)$ where $t \in \mathbb{Q}$. Then

$$t = \frac{Y}{X+1}, X^2 + Y^2 = 1$$

$$X^2 + t^2(X+1)^2 = 1 \implies (X+1)(t^2(X+1) + t^2 - 1) = 0$$

$$X = \frac{1-t^2}{1+t^2}$$

Thus,

$$t \mapsto \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right)$$

We will call this a "birational map". To find all integer solutions to $x^2 + y^2 = z^2$. Write $t = \frac{m}{n}$ then we see that

$$(x, y, z) = (n^2 - m^2, 2mn, n^2 + m^2)$$

is the solution to the Pythagorean equation.

Example (Cubic equations). We will study the elliptic curve with non-singular point like $y^2 = x^3 - 3x + 3$.

Go back to $x^2 + y^2 = 1$. This is a group since $e^{i(\theta_1 + \theta_2)} = e^{i\theta_1} e^{i\theta_2}$. It's also an algebraic variety. For $(x_1, y_1), (x_2, y_2) \in \mathbb{T}$, $(x_1, y_1) * (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$ which comes from sine and cosine.

1 Affine varieties

Definition 1.1. The affine space over k , defined by $\mathbb{A}^n(k)$ is

$$\mathbb{A}^n(k) := \{(x_1, \dots, x_n) \in k^n\}$$

Example. If $k = \mathbb{R}$, $\mathbb{A}^1(\mathbb{R})$ is the line, $\mathbb{A}^2(\mathbb{R})$ is the real plane.

Let $k[x_1, \dots, x_n]$ be the polynomial of n variables over k where

$$k[x_1, \dots, x_n] = \left\{ \sum_{(\alpha_1, \dots, \alpha_n) \in \mathbb{Z}^n} a_{\alpha_1, \dots, \alpha_n} x_1^{\alpha_1} \cdots x_n^{\alpha_n}, \quad a_{\alpha_1, \dots, \alpha_n} \in k \right\}$$

Example. $y^2 = x^3$ is a polynomial in $k[x, y]$.

Elements of $k[x_1, \dots, x_n]$ define functions on $\mathbb{A}^n(k)$. If $f \in k[x_1, \dots, x_n]$, $(a_1, \dots, a_n) \in \mathbb{A}^n(k)$ then $f(a_1, \dots, a_n)$ is a well-defined element of k .

Definition 1.2. Let $S \subseteq k[x_1, \dots, x_n]$ be a subset. Define $V(S) = \{(a_1, \dots, a_n) \in \mathbb{A}^n(k), f(a_1, \dots, a_n) = 0 \text{ for all } f \in S\}$. A subset of $\mathbb{A}^n(k)$ is called affine variety if it's of the form $V(S)$ for some $S \subseteq k[x_1, \dots, x_n]$.

Examples.

1. $\emptyset$ is an affine variety since $\emptyset = V(\{1\})$.
2. $V(S)$ is empty enough if $1 \notin S$. This happens if field is not algebraically closed. For example, $\mathbb{A}^2(\mathbb{R}) \supsetneq V(x^2 + y^2 + 1) = \emptyset$
3. $\mathbb{A}^n(k)$ itself is an affine variety since $\mathbb{A}^n(k) = V(\{0\})$.
4. $\mathbb{T}$ is an affine variety, it is $V(x^2 + y^2 - 1) = 0$.
5. A line is affine variety, for example $V(y - x)$.

Remark (Immediate Properties&exercises).

1. If $S_1 \subseteq S_2$ then $V(S_2) \subseteq V(S_1)$.
2. Let $S \subseteq k[x_1, \dots, x_n]$ be a subset. Let I be the ideal of $k[x_1, \dots, x_n]$ generated by S . Recall that

$$I = \left\{ \sum_{\text{finite}} a_i s_i, \quad a_i \in k[x_1, \dots, x_n], s_i \in S \right\}$$

then $V(I) = V(S)$.

3. It's possible that $V(S_1) = V(S_2)$ while $S_1 \neq S_2$ even if k is algebraically closed. In $\mathbb{A}^1(k)$, $V(x) = \{0\} = V(x^2)$.
4. Any point of $\mathbb{A}^n(k)$ is an affine variety. Indeed,

$$P = (a_1, \dots, a_n) = V(x_1 - a_1, \dots, x_n - a_n).$$

5. Affine subvarieties of $\mathbb{A}^1(k)$ are $\emptyset, \mathbb{A}^1(k)$. Others are of the form $V(I)$ where $I \in k[x]$ ideal. $I = (f)$ is principal, $V(I) = V(f)$.
6. Let I_1, I_2 be two ideals then $I_1 + I_2 = \{a_1 + a_2 : a_1 \in I_1, a_2 \in I_2\}$ is an ideal then $V(I_1 + I_2) = V(I_1) \cap V(I_2)$. In particular, the intersection of 2 affine varieties is an affine variety. In fact, if $\{I_i\}_{i \in \Lambda}$ is any collection of ideals then

$$V\left(\sum_{i \in \Lambda} I_i\right) = \bigcap_{i \in \Lambda} V(I_i).$$

Lemma 1.3. $V(I_1 \cap I_2) = V(I_1) \cup V(I_2)$ true for finite unions of affine varieties.

Proof. Let $P \in V(I_1) \cup V(I_2)$. WLOG we may assume $P \in V(I_1)$ then clearly $P \in V(I_1 \cap I_2)$ since $I_1 \cap I_2 \subseteq I_1$ so $V(I_1) \cup V(I_2) \subseteq V(I_1 \cap I_2)$. Now suppose $P \in V(I_1 \cap I_2)$ and $P \notin V(I_1) \cup V(I_2)$. Then there exists $f \in I_1$ and $g \in I_2$ such that $f(P)g(P) \neq 0$. Thus, this implies $P \notin V(I_1 \cap I_2)$ a contradiction. $\square$

Remark. $V(I_1 \cap I_2) = V(I_1) \cup V(I_2) = V(I_1 I_2)$.

Remark. If k is finite, then any subset $\mathbb{A}^n(k)$ is affine variety. Since $\emptyset, \mathbb{A}^n(k)$ are always affine variety. A finite set of points is always affine variety since a point $(a_1, \dots, a_n)$ is affine variety. A finite set is a finite union of point.

Remark. Any proper infinite subset is not affine variety.

Example. Open disk in $\mathbb{A}^2(\mathbb{R}) = \{(x, y) : x^2 + y^2 < 1\}$ is not affine variety. The intersection of two affine varieties is affine variety, If the disk is affine variety, then $\mathbb{A}^2(\mathbb{R}) \cap V(y)$ would be affine variety. But it's not since it's infinite.

Theorem 1.4 (Hilbert Basis Theorem). Any ideal in $k[x_1, \dots, x_n]$ is finitely generated. (in other words, any ideal I is of the form $I = (f_1, \dots, f_n)$, $f_i \in k[x_1, \dots, x_n]$). Equivalently, any ascending chain of ideals $I_1 \subseteq I_2 \subseteq \dots$ is stationary i.e. there exists N such that for all $n > N$, $I_n = I_N$.

If $V = V(I) = V(f_1, \dots, f_m)$ = set of solutions of the system

$$\begin{cases} f_1(x_1, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, \dots, x_n) = 0 \end{cases}$$

Note that $V(I) = V(f_1, \dots, f_m) = \bigcap_{i=1}^m V(f_i)$.

Definition 1.5. An affine variety of the form $V(f)$ is called a hypersurface.

Remark. If $\deg(f) = 1$ then $V(f)$ is called a hyperplane.

Then any affine variety is an intersection of finitely many hypersurfaces.

Example. In $\mathbb{A}^3(\mathbb{R})$, if we intersect $V(x^2 + y^2 - z^2)$ and $V(z - 1)$ then it's $V(x^2 + y^2 - z^2, z - 1)$. This is conic intersecting with hyperplane.

Remark. Sometimes affine variety can be defined by functions which are not algebraic.

Example. $\mathbb{A}^2(\mathbb{R}) \supset \{(\cos \theta, \sin \theta), \theta \in \mathbb{R}\} = V(x^2 + y^2 - 1)$ is an affine variety.

Example. $\mathbb{A}^2(\mathbb{R}) \supset \{(r, \theta) : r = \sin \theta\}$. Then $x^2 + y^2 = r^2 = r \sin \theta = y$ which implies the set is an affine variety.

Example. $\mathbb{A}^2(\mathbb{R}) \supset \{(r, \theta) : r \sin 2\theta\}$ which is an affine variety again but more complicated.

Example. $\{(r, \theta) : r = \sin \alpha \theta\}$ where $\alpha \in \mathbb{Q}$ is not an affine variety.

Example. $\{(x, y) : y = e^x\} \subset \mathbb{A}^2(\mathbb{R})$ is not an affine variety even if it has finite intersections with a line.

Proposition 1.6. Let $V \subset \mathbb{A}^n(k)$, $W \subset \mathbb{A}^m(k)$ be affine varieties. Let $V \times W = \{(x, y) : x \in V, y \in W\} \subset \mathbb{A}^{n+m}(k)$. Then $V \times W$ is an affine variety in $\mathbb{A}^{n+m}(k)$. More precisely, if $V = V(I)$ and $W = V(J)$ then $V \times W = V(K)$ where K is the ideal of $k[x_1, \dots, x_n, y_1, \dots, y_m]$ which is generated by I and J . In other words,

$$K = Ik[x_1, \dots, x_n, y_1, \dots, y_m] + Jk[x_1, \dots, x_n, y_1, \dots, y_m]$$

Proof. Consider two projections $\pi_1 : \mathbb{A}^{n+m} \rightarrow \mathbb{A}^n$ and $\pi_2 : \mathbb{A}^{n+m} \rightarrow \mathbb{A}^m$. We have

$$\mathbb{A}^{n+m} \supset \pi_1^{-1}(V(I)) = \{(a_1, \dots, a_n, b_1, \dots, b_m) : f(a_1, \dots, a_n) = 0, \forall f \in I\} = V(Ik[x_1, \dots, x_n, y_1, \dots, y_m])$$

Similarly,

$$\pi_2^{-1}(V(J)) = V(Jk[x_1, \dots, x_n, y_1, \dots, y_m])$$

Then $V \times W = V(I) \times V(J) = \{P = (a_1, \dots, a_n, b_1, \dots, b_m) \in \mathbb{A}^{n+m} : \pi_1(P) \in V, \pi_2(P) \in W = \pi_1^{-1}(V) \cap \pi_2^{-1}(W) = V(Ik[x_1, \dots, x_n, y_1, \dots, y_m]) \cap V(Jk[x_1, \dots, x_n, y_1, \dots, y_m])\}$. Recall that $V(I_1 + I_2) = V(I_1) \cap V(I_2)$ then we are done. $\square$

Example. Let $\mathbb{A}^1 \supset V = \{0\} = V(x)$ and $\mathbb{A}^1 \supset W = \{0\} = V(y)$. Then $\mathbb{A}^2 \supset V \times W = V(xk[x, y] + yk[x, y]) = V(x, y)$.

Example. Let $I = (x) \subset k[x]$ and $J = (y) \in k[y]$ then $K = Ik[x, y] + Jk[x, y] = (x, y) \subset k[x, y]$.

Definition 1.7. Let X be a subset of $\mathbb{A}^n(k)$. Let $I(X) = \{f \in k[x_1, \dots, x_n] : f(P) = 0, \forall P \in X\}$

Remark (Properties of $I(X)$).

1. $I(X)$ is an ideal.
2. If $X \subset Y$ then $I(Y) \subset I(X)$.
3. For any $X \in \mathbb{A}^n(k)$ then $X \subset V(I(X))$.
4. For any $J \subset k[x_1, \dots, x_n]$ then $J \subset I(V(J))$.
5. $I(\bigcup_{i \in I} X_i) = \bigcap_{i \in I} I(X_i)$.

Example. Let $J = (x^2) \subset k[x]$ then $V(J) = \{0\} \in \mathbb{A}^1$. While $I(V(J)) = (x) \subset k[x]$. Since $f(0) = 0$ if and only if $x \mid f$.

Example. Take $k = \mathbb{R}$ and $J = (x^2 + 1) \subsetneq k[x]$ and $V(J) = \emptyset$. Then $I(V(J)) = (1) = \mathbb{R}[x]$.

Proposition 1.8. Let V be an affine variety in $\mathbb{A}^n(k)$. Then $V(I(V)) = V$.

Proof. We already know that $V \subset V(I(V))$. Conversely, if $V = V(I)$ then $I \subset I(V)$ which implies $V(I) \supset V(I(V))$. $\square$

Proposition 1.9. If J is of the form $I(X)$ where X is affine variety. Then $J = I(V(J))$.

Generalising Euclidean division: Let R be a ring, let $f, g \in R[x]$ and $\deg f \geq \deg g$. We assume that the leading coefficient of g is a unit in R . Then there exists unique (q, r) such that $f = gq + r$ where $\deg(r) < \deg(g)$. For us, $k[x_1, \dots, x_n] = \underbrace{k[x_1, \dots, x_{n-1}]}_R[x_n]$ for instance $k[x, y] = k[x][y] = k[y][x]$.

Example. $I(\emptyset) = (1)$.

Example. Let k be any infinite field and $X = V(y - x^2)$. Clearly $(y - x^2) \subset I(X)$. Let $f \in I(X)$ then $f = (y - x^2)q(x, y) + r(x)$. Let $(a, a^2) \in X$ be any point on X for all $a \in k$, $f(a, a^2) = 0 = 0 + r(a)$ i.e. $r(a) = 0$ for all $a \in k$.

Example. Let k be any infinite field, and $X = V(y^2 - x^3)$. Consider the bijective map

$$t \mapsto (t^3, t^2)$$

Let $f \in I(X)$, we view $k[x, y]$ as $k[x][y]$, $y^2 - x^3$ is monic, we can divide by it:

$$f(x, y) = (y^2 - x^3)q + r_1(x)y + r_2(x)$$

For any $f \in I(X)$, for any $t \in k$,

$$f(t^3, t^2) = 0 = r_1(t^3)t^2 + r_2(t^3)$$

Let $r_1(x) = \sum_{i=0}^n a_i x^i$, $r_2(x) = \sum_{i=0}^m b_i x^i$. Then

$$\sum_{i=1}^n a_i t^{3i+2} + \sum_{i=0}^m b_i t^{3i} = 0$$

thus, $a_i, b_i = 0$ for all i . This yields $f = (y^2 - x^3)q \in (y^2 - x^3)$ i.e. $I(X) = (y^2 - x^3)$.

Definition 1.10 (Irreducible affine variety). Let V be a non-empty affine variety, V is irreducible if V cannot be written as $V_1 \cup V_2$ where $V_1, V_2 \neq V$.

Remark. $\emptyset$ is not irreducible.

Example.

1. $P = (a_1, \dots, a_n)$ is irreducible.
2. The union of two points is not irreducible.
3. If k is infinite, $\mathbb{A}^n(k)$ is irreducible.

Definition 1.11. Let R be a ring, an ideal $I \subset R$ is prime if for any $f, g \in R$ such that if $fg \in I$ then $f \in I$ or $g \in I$. Equivalently, R/I is an integral domain.

Theorem 1.12. Let $V \subset \mathbb{A}^n(k)$ be an affine variety then V is irreducible if and only if $I(V)$ is a prime ideal.

Proof. ($\Leftarrow$) : Suppose $I(V)$ is prime while V is reducible. Suppose, $V = V_1 \cup V_2$ where $V_1 \neq V \neq V_2$. This implies there exists some $f \in I(V_1)$ such that $f(P) = 0$ for some $P \in V$. Similarly, there exists some $g \in I(V_2)$ such that $g \notin I(V)$. Note that $fg \in I(V)$ but $f, g \notin I(V)$ a contradiction to the fact that $I(V)$ is prime.

($\Rightarrow$) : Suppose V is irreducible while V is not prime. There exists some $f, g \notin I(V)$ such that $fg \in I(V)$. Let $J_1 = (I(V), f)$ and $J_2 = (I(V), g)$. Notice that $V(J_1) \subset V$ and $V(J_2) \subset V$. As $fg \in I(V)$, we have $V = V_1 \cup V_2$. $\square$

Example. For $P = (a_1, \dots, a_n) \in \mathbb{A}^n(k)$, $I(P) = (x_1 - a_1, \dots, x_n - a_n)$ is the maximal ideal.

Example. Let k be infinite, consider $V = V(y^2 - x^3) = \{(t^2, t^3) : t \in k\}$. $f = 0$ on V and

$$f = (y^2 - x^3)q + r_1(x)y + r_2(x).$$

Plug in (t^2, t^3) we get

$$r_1(t^3)t^2 + r_2(t^3) = 0, \quad \forall t \in k.$$

then $r_1 = r_2 = 0$.

Remark. If k is finite, $\mathbb{A}^n(k)$ is a finite collection of points. If k is infinite, then $I(\mathbb{A}^n(k)) = (0)$ i.e. $\mathbb{A}^n(k)$ is irreducible.

Theorem 1.13. Any non-empty $V \subset \mathbb{A}^n(k)$ affine variety can be written as $V = V_1 \cup \dots \cup V_r$ where V_i 's are irreducible. This decomposition is unique up to permutation of one assumes that $V_i \not\subseteq V_j$ for all i, j . V_i 's are called irreducible components of V .

Proof. Recall Hilbert's Basis Theorem, any chain of ideals $I_1 \subset I_2 \subset \dots$ is stationary i.e. there exists some N such that $I_n = I_N$ for all $n \geq N$. Applying "V" to a chain $I_1 \subset I_2 \subset \dots$ we see that any descending chain of affine varieties in $\mathbb{A}^n(k)$:

$$V_1 \supset V_2 \supset \dots$$

will terminate i.e. there exists some N such that $V_n = V_N$ for all $n \geq N$.

(Existence): Let V be a non-empty affine variety. If V is irreducible, we are done. Otherwise, $V = V_1 \cup V_2$ where $V_1, V_2 \subsetneq V$. If V_1 is irreducible, then $V_1 = V'_1 \cup V'_2$ for $V'_1, V'_2 \subsetneq V_1$. Continue this process to construct descending chains of subvarieties of V . By Hilbert's Basis Theorem,

$$V = V_1 \cup \dots \cup V_r$$

And we assume that $V_i \not\subseteq V_j$ for any i, j .

(Uniqueness): Suppose we have

$$V = V_1 \cup \dots \cup V_r = V'_1 \cup \dots \cup V'_s$$

Let $j \in \{1, \dots, r\}$ then

$$V_j \subset V = \bigcup_{i=1}^s V'_i$$

and

$$V_j = V_j \cap V = \bigcup_{i=1}^s V_j \cap V'_i$$

so $V_j \subset V'_i$ for some i . Now similarly, there exists some j' such that $V'_i \subset V_{j'}$. Thus, $V_j = V'_i = V_{j'}$. Each V_j is equal to some V'_i which completes the proof of uniqueness. $\square$

Example. If $V = P_1 \cup P_2$ where $P_1 \neq P_2$ are two points then P_1, P_2 are irreducible components of V .

Example. For $k = \mathbb{R}$, $V(x^2 - y^2) = V((x - y)(x + y)) = V(x - y) \cup V(x + y)$ which is a decomposition into irreducible component.

Proposition 1.14 (Eisenstein Criterion). Let R be UFD, let

$$f = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0.$$

Suppose there is a prime element (equivalently irreducible) p such that $p \mid a_i$ for $0 \leq i \leq n - 1$ but $p^2 \nmid a_0$ then f is irreducible in $R[x]$.

Remark. Condition $p^2 \nmid a_0$ is necessary for example $x^2 - 4$ is not irreducible in $\mathbb{Z}[x]$.

Theorem 1.15 (Zariski's Lemma). Let K be a finitely generated algebra over k . Assume K is a field then $k \subset K$ is a finite field extension.

Remark. If k is algebraically closed. Then in the conclusion of Zariski's Lemma, $K = k$. Any ideal of the form $(x_1 - a_1, \dots, x_n - a_n) = m$ is maximal i.e. $k[x_1, \dots, x_n]/m \cong k$. The converse is false for a general k for example if $k = \mathbb{R}$, $m = (x^2 + 1)$ then $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.

Remark. Zariski's Lemma implies that if k is algebraically closed, then any maximal ideal in $k[x_1, \dots, x_n]$ is of the form $(x_1 - a_1, \dots, x_n - a_n)$. Let $m \subset k[x_1, \dots, x_n]$ be a maximal ideal. If k is algebraically closed then $k = k[x_1, \dots, x_n]/m$.

Theorem 1.16 (Nullstellensatz). Let k be algebraically closed, $I \subset k[x_1, \dots, x_n]$ an ideal. If $V(I) = \emptyset$ then $1 \in I$. Equivalently, if $I \neq k[x_1, \dots, x_n]$ then $V(I) \neq \emptyset$.

Proof. Let $I \neq k[x_1, \dots, x_n]$ then $I \subset m$ is contained in some maximal ideal. Since k is algebraically closed, by Zariski's Lemma, $m = (x_1 - a_1, \dots, x_n - a_n)$. If $I \subset m$ then $V(m) \subset V(I)$ so clearly $V(I) \neq \emptyset$. $\square$

Remark. By Hilbert Basis Theorem, $I = (f_1, \dots, f_m)$ so

$$V(I) = \left\{ \begin{array}{l} f_1(x_1, \dots, x_n) = 0 \\ \vdots \\ f_m(x_1, \dots, x_n) = 0 \end{array} \right.$$

has a solution if there is no relation of form $1 = \sum_i^n g_i f_i$.

When is $I = I(V(I))$? There are at least 2 obstructions:

1. If k is not algebraically closed, then $V(I)$ may be empty even if $1 \notin I$. Take $k = \mathbb{R}$, $I = (x^2 + 1)$ and $V(I) = \emptyset$.
2. I may contain elements f^m without containing f itself. Let

$$I = (x^2) \subset \mathbb{C}[x]$$

but $I(V(I)) = (x)$.

Definition 1.17 (Radical). Let $I \subset R$ be an ideal of a ring, the radical of ideal is defined via

$$\sqrt{I} = \{a \in R \mid \exists r \in \mathbb{N}, a^r \in I\}$$

I is called radical if $\text{rad}(I) = I$.

Remark. Any prime ideal is radical.

Remark. If V is an affine variety then $I(V)$ is radical. Indeed, $I(V) = \{f : f(P) = 0, \forall P \in V\}$. If $f^n \in I(V)$ then $(f(P))^n = 0$ for all $P \in V$ so $f(P) = 0$ for all $P \in V$ i.e. $f \in I(V)$.

Proposition 1.18. $\sqrt{I}$ is an ideal of R .

Remark. One can show that $\sqrt{I} = \bigcap_p \text{prime in } R \text{ containing } I$

Example. If $I = (x^2)$ in $k[x]$ then $\sqrt{I} = (x)$. Similarly, if $I = (x^2, y)$ then $(x, y) \in \sqrt{I}$. Another example is $I = (x^2, x^4 - x) \in k[x]$, then $x \in I$ so I is radical.

Theorem 1.19 (Strong Nullstellensatz). Let k be algebraically closed, and V be an affine variety then

$$I(V(I)) = \sqrt{I}$$

Proof. This theorem is equivalent to "Weak Nullstellensatz".

($\implies$) : Assume $V(I) = \emptyset$. By S.N, $1 \in \sqrt{I}$, then $1^n \in I$ for some n i.e. $1 \in I$.

($\impliedby$) : We want to find $I(V(I))$. Let $f \in I(V(I)) \subseteq k[x_1, \dots, x_n] \subseteq k[x_1, \dots, x_n, y]$. Let $J = (I, fy - 1)$ be the ideal generated by I and $f(x_1, \dots, x_n)y - 1$. For $\emptyset \neq V(J) \subset \mathbb{A}^{n+1}(k)$, $P = (a_1, \dots, a_{n+1}) \in V(J)$ if

$$g(a_1, \dots, a_n) = 0, \quad \forall g \in I$$

$$f(a_1, \dots, a_n)a_{n+1} = 1$$

If $(a_1, \dots, a_n) \in V(I)$ which exists by W.N. Then this shows $0 = 1$ which is impossible i.e. $V(J) = \emptyset$. Therefore, by W.N., $1 \in J$:

$$1 = \sum g_i h_i + g_0(f(x_1, \dots, x_n)y - 1), \quad g_i \in k[x_1, \dots, x_n, y], h_i \in I \subset k[x_1, \dots, x_n], g_0 \in k[x_1, \dots, x_n, y]$$

Substitute $y = \frac{1}{f(x_1, \dots, x_n)}$ we get

$$1 = \sum g_i(x_1, \dots, x_n, \frac{1}{f(x_1, \dots, x_n)}) h_i(x_1, \dots, x_n)$$

This implies

$$f^n = \sum g'_i(x_1, \dots, x_n) h_i(x_1, \dots, x_n) \in I$$

i.e. $f \in \sqrt{I}$. $\square$

Definition 1.20. The topological space (X, τ) where X is a set and τ is a collection of subsets of X such that:

1. $\emptyset, X \in \tau$.
2. Finite intersection of elements of τ is in τ .
3. Arbitrary union of elements of τ is in τ .

Elements of τ are called open subsets of X . Their complements are called closed subsets of X .

Example. Trivial topology: take any X , then $\tau = \{\emptyset, X\}$ is the trivial topology.

Definition 1.21 (Basis of topology). A basis of a topology is $\Sigma \subset \tau$ such that

1. Σ covers X i.e. for all $x \in X$, there exists $U \in \Sigma$ such that $x \in U$.
2. If $U, V \in \Sigma$, there exists $W \in \Sigma$, $W \subset U \cap V$.

Example. Discrete topology: X any set, $\tau = \{\text{All open subsets of } X\}$. For discrete topology, any subset of X is clopen (closed and open).

Definition 1.22 (Hausdorff property). A topological space (X, τ) is said to have Hausdorff property if for any $x, y \in X$, $x \neq y$, there exists $U \ni x$ and $V \ni y$ open such that $U \cap V = \emptyset$.

Example. The usual topology on $\mathbb{R}$ (or even metric space) has Hausdorff topology.

Definition 1.23 (Zariski topology). The Zariski topology on $\mathbb{A}^n(k)$ is the one in which closed sets are the affine varieties i.e. $V(I)$, $I \subset k[x_1, \dots, x_n]$.

Remark. This satisfies the axioms, so it is a topology.

Definition 1.24. The open sets in Zariski topology is defined via

$$D(I) = V(I)^c = \{P : \exists f \in I, f(P) \neq 0\}$$

Remark. Any $I = (f_1, \dots, f_m)$ finitely generated, $V(I) = V(f_1) \cap \dots \cap V(f_m)$. Then

$$D(I) = D(f_1) \cup \dots \cup D(f_m).$$

The sets $D(f)$ are called basis open sets.

Lemma 1.25. Any two (Zariski) non-empty open sets intersect. In particular, Zariski topology is not Hausdorff.

Proof. Assume $D(f) \cap D(g) = \emptyset$ then $V(f) \cup V(g) = \mathbb{A}^n(k)$ i.e. $V(fg) = \mathbb{A}^n(k)$. This implies $fg = 0$ on $\mathbb{A}^n(k)$ so $f = 0$ or $g = 0$. Thus, either $D(f)$ or $D(g)$ is $\emptyset$. $\square$

Definition 1.26. Let X be any topological space, $A \subset X$ be a subset. The closure of A denoted $\bar{A}$ is the smallest closed set containing A . A is called dense if $\bar{A} = X$.

Exercise. In Zariski topology, any non-empty open set is dense.

Definition 1.27. Let X, Y be two topological spaces. Then product topology on the direct product $X \times Y$, is the one where open sets are unions of the $U \times V$ where U is open in X and V is open in Y .

Remark. The Zariski topology on a product is not the product topology.

Remark. Affine variety is equivalent to algebraic subset of $\mathbb{A}^n(k)$ which is closed subset of $\mathbb{A}^n$.

Definition 1.28. Let X be any topological space, $Y \subset X$ a subset. Y can be endowed with a topology by saying that a closed subset of Y is the intersection of a closed set in X with Y .

Remark. In our case: $X = \mathbb{A}^n(k)$ with Zariski topology and Y is an affine variety of $\mathbb{A}^n(k)$. Y has an induced Zariski topology in which closed sets are intersections of Y with $V(I) \subseteq \mathbb{A}^n(k)$.

Remark. Closed subsets of Y are closed in X since they are intersections of closed sets in X .

Remark. Open sets of Y are necessarily open in X . Trivial example is Y itself.

Example. Here's a non-trivial example. Take $X = \mathbb{A}^2(k)$, $Y = \mathbb{A}^1(k) \subset \mathbb{A}^2(k)$. Consider $\{(x, 0) \mid x \neq 0\}$, but it's not open in X .

Definition 1.29. A topological space X is called irreducible if X cannot be written as $X = X_1 \cup X_2$ where $X_1, X_2 \neq X$ closed.

Remark. This definition is useless in the usual topology on $\mathbb{R}$ since $\mathbb{R}$ itself is not irreducible.

Definition 1.30. A topological space X is connected if X cannot be written as $X = X_1 \cup X_2$ where $X_1, X_2 \neq X$ closed and $X_1 \cap X_2 = \emptyset$.

Remark. In any topological space, irreducible implies connected.

Example. $\mathbb{A}^1(k) \supset \{0\} \cup \{1\}$ not connected. And $V((y - x^2)(x - y)) = V(y - x) \cup V(x - y)$ is connected but not irreducible. In addition, $V(xy - 1)$ is irreducible and connected in Zariski topology.

Definition 1.31. Let k be an infinite field, any $f \in k[x_1, \dots, x_n]$ defines a function on $\mathbb{A}^n(k)$ and by restriction on any $V \subset \mathbb{A}^n(k)$, an affine variety. So f defines a function $V \xrightarrow{f} k$ called a polynomial (or regular function).

Example. Each x_i defines a regular function on $\mathbb{A}^n$ given by

$$(a_1, \dots, a_n) \mapsto a_i.$$

Let $\mathcal{F}(V, k) = \{\text{All functions on } V\}$ which is a ring:

$$\begin{aligned} k[x_1, \dots, x_n] &\xrightarrow{\lambda} \mathcal{F}(V, k) \\ f &\mapsto \text{The function defined by } f|_V \end{aligned}$$

and λ is a ring homomorphism and

$$\ker(\lambda) = \{f \in k[x_1, \dots, x_n], f = 0 \text{ on } V\} = I(V).$$

We get an injection

$$\Gamma(V) = k[x_1, \dots, x_n]/I(V) \hookrightarrow \mathcal{F}(V, k)$$

is called coordinate ring of V .

Example. Consider $V = V(y - x^2) \subset \mathbb{A}^2$ then $\Gamma(V) = k[x, y]/(y - x^2)$ generated (as k -algebra) by 2 functions $\bar{x}, \bar{y}$.

Definition 1.32. A k -algebra is a ring R which has a structure of k -vector space compatible with ring operation. A subset $S \subset R$ generates R as k -algebra if

$$R = \left\{ \sum_{I=\{a_1, \dots, a_n\}, n \geq 1} a_I s_1^{a_1} \cdots s_n^{a_n} \right\}$$

Finitely generated: S can be chosen to be finite.

Remark. In this course: We look at finitely generated algebras over fields k : They are simply $k[x_1, \dots, x_n]/I$. The coordinate ring $\Gamma(V) = k[x_1, \dots, x_n]/I(V)$ is a finitely generated algebra over k .

Example. Consider

$$\begin{aligned} \mathbb{A}^1(k) &\rightarrow k \\ x &\mapsto x^2 \end{aligned}$$

then $\Gamma(\mathbb{A}^1(k)) = k[x]$ because $I(\mathbb{A}^1(k)) = (0)$. More generally: $\Gamma(\mathbb{A}^n(k)) = k[x_1, \dots, x_n]$.

Let $f_1, \dots, f_m \in k[x_1, \dots, x_n]$. These define a map between

$$\mathbb{A}^n \rightarrow \mathbb{A}^m$$

$$(a_1, \dots, a_n) \mapsto (f_1(a_1, \dots, a_n), \dots, f_m(a_1, \dots, a_n))$$

Let $V \subset \mathbb{A}^n$, $W \subset \mathbb{A}^m$ be affine varieties then a polynomial map (also called a morphism) between V and W is

$$\begin{aligned} V &\rightarrow W \\ (a_1, \dots, a_n) &\mapsto (f_1(a_1, \dots, a_n), \dots, f_m(a_1, \dots, a_n)) \end{aligned}$$

Let $V \xrightarrow{\varphi} W \xrightarrow{f} k$ be a polynomial map, and f be a polynomial function $W \rightarrow k$. Then $f \circ \varphi = \varphi^*(f)$ (also called the pullback) is a polynomial map $V \rightarrow k$. We get a map $\Gamma(w) \xrightarrow{\varphi^*} \Gamma(v)$ by sending f to $\varphi^*(f) = f \circ \varphi$. If we have V, W, Z varieties and $V \xrightarrow{\varphi} W$ and $W \xrightarrow{\psi} Z$ we have $V \xrightarrow{\psi \circ \varphi} Z$. Then $\psi \circ \varphi = \tilde{\varphi} \circ \tilde{\psi} : \Gamma(Z) \rightarrow \Gamma(V)$. If V and W are isomorphic, then $\Gamma(V)$ and $\Gamma(W)$ are isomorphic as k -algebras.

Exercise. $\tilde{\varphi}$ is a k -algebra homomorphism.

Definition 1.33. Let X, Y be topological spaces. $f : X \rightarrow Y$ a map is said to be continuous if for any $F \subset Y$ open, $f^{-1}(F)$ is open in X .

Example. $V = V(xy - 1)$ is not isomorphic to $\mathbb{A}^1$. Since $\Gamma(V) = k[x, y]/(xy - 1) = k[x, x^{-1}]$ while $\Gamma(\mathbb{A}^1) = k[x]$ so $k[x]$ and $k[x, x^{-1}]$ are not isomorphic since $(k[x])^* = k^*$ but $k[x, x^{-1}]^* \ni x$ which is not in k .

Conversely: Let $\mathbb{A}^n \supset V$, $\mathbb{A}^m \supset W$. Let $\alpha : \Gamma(W) \rightarrow \Gamma(V)$ be a k -algebra morphism. We would like to construct $\varphi : V \rightarrow W$ polynomial such that $\tilde{\varphi} = \alpha$. Consider

$$k[x_1, \dots, x_n]/I(W) = \Gamma(W) \xrightarrow{\alpha} \Gamma(V) = k[x_1, \dots, x_n]/I(V)$$

α corresponds to

$$\begin{aligned} \tilde{\alpha} : k[x_1, \dots, x_m] &\rightarrow k[x_1, \dots, x_n] \\ x_i &\mapsto f_i(x_1, \dots, x_n) \end{aligned}$$

This gives a polynomial map

$$\begin{aligned} \varphi : \mathbb{A}^n &\rightarrow \mathbb{A}^m \\ (a_1, \dots, a_n) &\mapsto (f_1(a_1, \dots, a_n), \dots, f_m(a_1, \dots, a_n)) \end{aligned}$$

We need to check $\varphi(V) \subset W$. Let $(a_1, \dots, a_n) \in V$, we need to show: for any $f \in I(W)$, $f(\varphi(a_1, \dots, a_n)) = 0$. Let $f \in I(W)$,

$f(\varphi(a_1, \dots, a_n)) = f(f_1(a_1, \dots, a_n), \dots, f_m(a_1, \dots, a_n)) = \tilde{\alpha}(f)(a_1, \dots, a_n) = 0$ since $f \in I(W)$, $\tilde{\alpha}(I(W)) \subset I(V)$. To show uniqueness, suppose we have $\varphi : V \rightarrow W$ and $\varphi' : V \rightarrow W$ such that $\tilde{\varphi} = \tilde{\varphi}' = \alpha$ we want to show that $\varphi = \varphi'$. Let $(a_1, \dots, a_n) \in V$ then

$$x_i \circ \varphi = \varphi^*(x_i)(a_1, \dots, a_n) = \alpha(x_i)(a_1, \dots, a_n) = f_i(a_1, \dots, a_n)$$

Similarly,

$$x_i \circ \varphi' = \varphi'^*(x_i)(a_1, \dots, a_n) = \alpha(x_i)(a_1, \dots, a_n) = f_i(a_1, \dots, a_n)$$

i.e. $\varphi = \varphi'$.

Remark. For any k -algebra morphism, $\alpha : \Gamma(W) \rightarrow \Gamma(V)$, there is a unique polynomial map $\varphi : V \rightarrow W$ such that $\tilde{\varphi} = \alpha$.

Remark. Important Consequence: V and W are isomorphic if and only if $\Gamma(V)$ and $\Gamma(W)$ are isomorphic as k -algebra.

Example. Consider $V = V(y - x^2) \subset \mathbb{A}^2$. Then

$$\begin{aligned} \varphi : V &\rightarrow \mathbb{A}^1 \\ (x, y) &\mapsto x. \end{aligned}$$

And $\Gamma(V) = k[x, y]/(y - x^2)$, $\Gamma(\mathbb{A}^1) = k[x]$. And

$$\begin{aligned} \tilde{\varphi} : k[x] &\rightarrow k[x, y]/(y - x^2) \\ x &\mapsto x \end{aligned}$$

which is an isomorphism.

Example. Suppose $k = \mathbb{R}$ consider

$$\begin{aligned} \mathbb{A}^1 &\xrightarrow{\varphi} V(y^3 - x^2) = V \\ t &\mapsto (t^3, t^2) \end{aligned}$$

φ is bijective but φ is not an isomorphism. $\Gamma(V) = k[x, y]/(y^3 - x^2) = k[t^2, t^3]$ while $\Gamma(\mathbb{A}^1) = k[t]$. Then

$$\tilde{\varphi} : k[t] \hookrightarrow k[t^2, t^3]$$

is not an isomorphism since $t \notin k[t^2, t^3]$ so φ is not an isomorphism despite being bijective.

Remark. In general, $k[t]$ is not isomorphic to $k[t^2, t^3]$ because $k[t]$ is UFD. But $k[t^2, t^3]$ is not UFD since $(t^2)^3 = (t^3)^2$. On can check that t^2, t^3 are irreducible in $k[t^2, t^3]$.

Example. Consider

$$\begin{aligned} f : V = \mathbb{A}^1 &\rightarrow \mathbb{A}^2 = W \\ x &\mapsto (x, 0) \end{aligned}$$

then $k[x, y] = \Gamma(W) \rightarrow \Gamma(V) = k[x]$ is given by $g(x, y) \mapsto g(x, 0)$.

Example. Consider

$$\begin{aligned} \mathbb{A}^2 &\xrightarrow{f} \mathbb{A}^1 \xrightarrow{\varphi} k \\ (x, y) &\mapsto x \end{aligned}$$

$k[x] \hookrightarrow k[x, y]$ by $x \mapsto x$.

$$\begin{aligned} \mathbb{A}^1 &\rightarrow \mathbb{A}^1 \\ x &\mapsto \frac{1}{x} \end{aligned}$$

This is an example of a rational function which is not defined at $x = 0$ but it's well-defined on $\mathbb{A}^1 \setminus \{0\}$ which is a Zariski

open subset of $\mathbb{A}^1$. Let $V \subseteq \mathbb{A}^n$ be an irreducible variety. $\Gamma(V) = k[x_1, \dots, x_n]/I(V)$ is an integral ring. Let $k(V)$ be a field of fractions of $\Gamma(V)$.

Remark. Let R be an integral domain, and

$$\begin{aligned} K &= \left\{ \frac{f}{g} : f, g \in R, g \neq 0 \right\} \\ \frac{f}{g} &= \frac{f'}{g'} \implies fg' = f'g \\ \frac{f}{g} \cdot \frac{f'}{g'} &= \frac{ff'}{gg'} \\ \frac{f}{g} + \frac{f'}{g'} &= \frac{g'f + f'g}{gg'}. \end{aligned}$$

Let V be irreducible, let

$$k(V) = \left\{ \frac{f}{g} : f, g \in \Gamma(V), g \neq 0 \right\}$$

which is called field of rational functions on V .

Definition 1.34. Let $\varphi \in k(V)$. We say that φ is regular at x if $\varphi = \frac{f}{g}$ where $g(x) \neq 0$.

Remark. It's possible that $\varphi = \frac{f}{g}$ with $g(x) = 0$ but for φ still to be regular at x . For example, $\varphi = \frac{x}{x}$ is regular at $x = 0$.

If φ is not regular at x then x is called a pole of φ .

Example. $\varphi = \frac{1}{x}$ on $\mathbb{A}^1$ then φ is not regular at $x = 0$. Assume φ was regular at 0,

$$\frac{1}{x} = \frac{f}{g} \text{ with } g(0) \neq 0$$

then $g = x \cdot f$ in $k[x]$. Evaluate at $x = 0$ we have $g(0) = 0$, a contradiction.

If $\varphi \in k(V)$ is regular at x , $\varphi = \frac{f}{g}$, $x \neq 0$ then the value of φ at x is well-defined: $\varphi(x) = \frac{f(x)}{g(x)}$.

Example. $V = V(xy - zt) \subseteq \mathbb{A}^4$ and $\Gamma(V) = k[x, y, z, t]/(xy - zt)$, $\varphi = \frac{x}{z} \in k(V)$. Where is φ regular? Clearly, φ is regular for $z \neq 0$. In $k(V)$, we have $xy = zt$ so

$$\varphi = \frac{x}{z} = \frac{t}{y} \text{ in } k(V).$$

φ is also regular when $y \neq 0$. φ is regular on $D(z) \cup D(y)$. We established that φ is regular outside $V(z, y) \subseteq V$. Can φ be regular anywhere on $V(z, y)$. Assume $\varphi = \frac{f}{g}$ with $g \neq 0$ on $V(z, y)$. In $k(V)$, $\frac{f}{g} = \frac{x}{z}$ then

$$zf = xg \text{ in } \Gamma(V).$$

i.e. $zf - xg \in (xy - zt)$ so $zf - xg = (xy - zt) \cdot h$ for some $h \in k[x, y, z, t]$. For $z = 0$,

$$-x \cdot g(x, 0, y, t) = xy \cdot h(x, 0, y, t)$$

$$-g(x, 0, y, t) = yh(x, 0, y, t) \implies g(a, 0, 0, b) = 0$$

We assumed that $g \neq 0$ on $V(z, y)$ which means $g(a, 0, 0, b) \neq 0$ for some a, b . But we have

$$g(a, 0, 0, b) = 0$$

a contradiction.

We showed that φ is regular exactly on $D(z) \cup D(y)$ which is a Zariski open set. The set of poles of φ is $V(z, y)$ which is Zariski closed.

Definition 1.35. Given φ , $\{x : \varphi \text{ is regular at } x\}$ is called the domain of definition of φ .

Proposition 1.36. The domain of definition of φ is Zariski open set. (Equivalently, $\{\text{poles of } \varphi\}$ is Zariski closed).

Proof. Look at all representations of φ as $\frac{f}{g}$. Let

$$\Sigma = \left\{ x : \text{all the } g' \text{ that come up as } \varphi = \frac{f}{g} \text{ with } g = 0 \right\} = \bigcap_{g \text{ occurs } \varphi = \frac{f}{g}} V(g) = \{\text{Poles of } \varphi\}$$

and we are done since the above is Zariski closed as intersection of Zariski closed sets. $\square$

$$\Gamma(V) \subset k(V)$$

$$f \mapsto \frac{f}{1}$$

Polynomial functions are regular everywhere on V . But the converse is not true for example

$$\varphi : \mathbb{A}^1 \rightarrow \mathbb{R}$$

$$x \mapsto \frac{1}{x^2 + 1}$$

Proposition 1.37. Assume k is algebraically closed. Let $\varphi \in k(V)$ regular everywhere. Then $\varphi \in \Gamma(V)$.

Proof. For all $x \in V$, $\exists f_x, g_x \in \Gamma(V)$, $g_x(x) \neq 0$ and $\varphi = \frac{f_x}{g_x}$. Let $I \subseteq \Gamma(V)$ be the ideal generated by the g_x , $x \in V$. By Hilbert's Basis Theorem, I is finitely generated:

$$I = (g_{x_1}, \dots, g_{x_n})$$

$V(I)$ is the set of poles of φ which is emptyset by assumption. Now as k is algebraically closed, by Weak Nullstellensatz Theorem, $1 \in I = (g_{x_1}, \dots, g_{x_n})$ i.e.

$$1 = h_1 g_{x_1} + \dots + h_n g_{x_n}$$

Then

$$\varphi = h_1(\varphi g_{x_1}) + \dots + h_n \varphi g_{x_n} = h_1 f_{x_1} + \dots + h_n f_{x_n} \in \Gamma(V).$$

□

Remark (Non-examinable). Let V be irreducible,

$$\{\text{Zariski Open Sets}\} \rightarrow \text{Rings}$$

$$U \rightarrow O_V(U) = \{\varphi \in k(V), \varphi \text{ regular on } U\}$$

And

$$O_V(U) \rightarrow O_V(W)$$

$$\varphi \mapsto \varphi|_W$$

This is called the structure sheaf of V . Elements of $O_V(U)$ are called sections over U . If k is algebraically closed, then $O_V(U) = \Gamma(U)$.

Let $V \subseteq \mathbb{A}^n$, $W \subseteq \mathbb{A}^m$ irreducible. A rational map $V \xrightarrow{\varphi} W$ is one of the form

$$(a_1, \dots, a_n) \mapsto (\varphi_1(a_1, \dots, a_n), \dots, \varphi_m(a_1, \dots, a_n)) = \varphi(a_1, \dots, a_n).$$

where $\varphi_i \in k(V)$.

Example.

$$\mathbb{A}^1 \xrightarrow{\varphi} \mathbb{A}^1$$

$$x \mapsto \frac{1}{x}$$

is a rational map. It is defined on $\mathbb{A}^1 \setminus \{0\}$.

Definition 1.38. $\{x : \varphi \text{ is defined at } x\}$ is called the domain of definition of φ

Example. Let $V = V(x^2 + y^2 - 1)$:

$$\varphi : V \rightarrow \mathbb{A}^1$$

$$(x, y) \mapsto \frac{y}{1-x}$$

is a rational function defined where $x \neq 1$. φ is defined on $V \setminus \{(1, 0)\}$.

Exercise. Show that φ is not regular at $(1, 0)$.

Example. Consider

$$\mathbb{A}^1 \xrightarrow{\psi} V$$

$$t \mapsto \left(\frac{t^2 - 1}{t^2 + 1}, \frac{2t}{t^2 + 1} \right)$$

is a rational function. If $k = \mathbb{R}$, ψ is regular everywhere (but not polynomial) since k is not algebraically closed.

Exercise. If $k = \mathbb{C}$, ψ is defined for $t \neq \pm i$. Show that ψ is not defined for $t = \pm i$.

Example. Let $V = V(y^2 - x^3)$,

$$\begin{aligned}\varphi : \mathbb{A}^1 &\rightarrow V \\ t &\mapsto (t^2, t^3)\end{aligned}$$

But it's "inverse":

$$\begin{aligned}V &\xrightarrow{\psi} \mathbb{A}^1 \\ (x, y) &\mapsto \frac{y}{x}\end{aligned}$$

is rational, defined when $x \neq 0$.

Exercise. Show that ψ is not defined at $(0, 0)$.

If we have polynomial maps $\varphi : V \rightarrow W$, $\psi : W \rightarrow T$. We have $\psi \circ \varphi : V \rightarrow T$ which is a polynomial map. Look at rational functions:

$$\begin{aligned}\mathbb{A}^2 &\xrightarrow{\varphi} \mathbb{A}^1 \\ (x, y) &\mapsto \frac{1}{1 - x^2 - y^2}\end{aligned}$$

It is not defined on $V(x^2 + y^2 - 1)$. If

$$\begin{aligned}\psi : \mathbb{A}^1 &\rightarrow \mathbb{A}^2 \\ t &\mapsto \left(\frac{t^2 - 1}{t^2 + 1}, \frac{2t}{t^2 + 1} \right)\end{aligned}$$

then $\varphi \circ \psi$ does not make sense because φ is not defined on $\text{Im}(\psi) = V(x^2 + y^2 - 1)$.

Definition 1.39. $\varphi : V \rightarrow W$ rational is called dominant if $\text{Im}(\varphi)$ is Zariski dense in W . That means $W = V(I(\text{Im}(\varphi)))$ i.e. for any $f \in \Gamma(W)$ which is zero on $\text{Im}(\varphi)$, is zero on W .

$$\begin{aligned}\mathbb{A}^1 &\xrightarrow{\varphi} \mathbb{A}^1 \\ x &\mapsto \frac{1}{x}\end{aligned}$$

is dominant: $\text{Im}(\varphi) = \mathbb{A}^1 \setminus \{0\}$ Zariski open in $\mathbb{A}^1$ hence dense. Let $\varphi : V \rightarrow W$ dominant, $\psi : W \rightarrow T$ any rational map then

$$\psi \circ \varphi : V \rightarrow T$$

is a well-defined rational map because the domain of definition of ψ is Zariski open in W and so intersects $\text{Im}(\varphi)$ because it's Zariski dense. If $\varphi : V \rightarrow W$ and $\psi : W \rightarrow V$ are both dominant we have: $\psi \circ \varphi : V \rightarrow V$, $\varphi \circ \psi : W \rightarrow W$ are rational map. φ and ψ are said to be rational inverses of each other if $\psi \circ \varphi = \text{id}_V$ and $\varphi \circ \psi = \text{id}_W$ where it makes sense. A rational function $\varphi : V \rightarrow W$ which has a rational inverse is called birational (also birational equivalence). If φ exists, V and W are called birational (or birational equivalence).

Example.

$$\begin{aligned}\mathbb{A}^1 &\xrightarrow{\varphi} V(y^2 - x^3) \\ t &\mapsto (t^2, t^3)\end{aligned}$$

is birational.

Example.

$$\begin{aligned}V(x^2 + y^2 - 1) &\xrightarrow{\varphi} \mathbb{A}^1 \\ (x, y) &\mapsto \frac{y}{x - 1}\end{aligned}$$

is birational.

Example (Elliptic Curve). $y^2 = x^3 + Ax + B$ where $\Delta = 4A^3 + 27B^2 \neq 0$ which is not birational to $\mathbb{A}^1$.

Suppose φ is dominant. For any $f \in \Gamma(W)$, $\varphi^*(f) = f \circ \varphi$ is a well-defined element of $k(V)$. This gives a field homomorphism

$$\begin{aligned}\varphi^* : k(W) &\rightarrow k(V) \\ f &\mapsto f \circ \varphi\end{aligned}$$

which is non-zero.

Exercise. Given a non-zero field homomorphism $\alpha : k(W) \rightarrow k(V)$, there is a unique $\varphi : V \rightarrow W$ rational map, which is dominant and $\varphi^* = \alpha$. In particular, this implies that V and W are birational iff $k(V) \cong k(W)$.

Example.

$$\begin{aligned}\varphi : \mathbb{A}^1 &\rightarrow \mathbb{A}^1 \\ x &\mapsto \frac{1}{x}\end{aligned}$$

and

$$\begin{aligned}\varphi^* : k(x) &\rightarrow k(x) \\ x &\mapsto f\left(\frac{1}{x}\right)\end{aligned}$$

is injective.

Remark. Suppose $\varphi : V \rightarrow W$ is not dominant. There is non-zero $f \in \Gamma(W)$ which is zero on $\varphi(V)$.

2 Projective varieties

Definition 2.1. $\mathbb{P}^n(k) = \{\text{Lines through } 0 \text{ in } \mathbb{A}^{n+1}\}$. A line through O is by definition $\{\lambda x, \lambda \in k\}$ where $x \neq 0$. So $(x_0, \dots, x_n)$ and $(y_0, \dots, y_n)$ represent the same line iff $\exists \lambda \in k \setminus \{0\}, (y_0, \dots, y_n) = (\lambda x_0, \dots, \lambda x_n)$.

This defined what is called homogeneous coordinates on $\mathbb{P}^n(k)$. $[x_0 : \dots : x_n]$ represents the line through O and $(x_0, \dots, x_n) \neq 0$.

for any $\lambda \neq 0$. Equivalently, we have $\mathbb{P}^n(k) = \mathbb{A}^{n+1} \setminus \{0\} / \sim$ where $(x_0, \dots, x_n) \sim (y_0, \dots, y_n)$ iff $\exists \lambda \neq 0$ such that $y_i = \lambda x_i$.

Example. For $n = 1$, $\mathbb{P}^1 = \{[a : b], a, b \in k, (a, b) \neq 0\}$. Let $[a : b] \in \mathbb{P}^1$, If $a \neq 0$, $[a : b] = [1 : \frac{b}{a}]$. If $b \neq 0$, $[a : b] = [\frac{a}{b} : 1]$.

Generally, if $[x_0 : \dots : x_n] \in \mathbb{P}^n$, you can always make one of the coordinates equal 1. Thus,

$$\mathbb{P}^n = \{[1 : x_1 : \dots : x_n]\} \cup \{[0 : x_1 : \dots : x_n]\}$$

Let $n = 1$,

$$\mathbb{P}^1 = \{[1 : x]\} \cup \{[0 : 1]\}$$

where $[0 : 1]$ is called the point of infinity.

Example. Let $n = 1$, $\mathbb{P}^1(\mathbb{R}) \cong S^1$.

Let $f \in k[x_0, \dots, x_n]$ we can always evaluate f at $(a_0, \dots, a_n) \in \mathbb{A}^{n+1}$. But we can not evaluate f at $[a_0 : \dots : a_n] \in \mathbb{P}^n$ because $f(\lambda a_0, \dots, \lambda a_n) \neq f(a_0, \dots, a_n)$ in general. Even saying that $f([a_0 : \dots : a_n]) = 0$ is meaningless.

Example. In $\mathbb{P}^1$, $f = y - x^2$ so $f(1, 1) = 0$ but $f(2, 2) \neq 0$.

Definition 2.2 (Homogeneous). Let

$$f = \sum_{(a_0, \dots, a_n)} \alpha_{a_0, \dots, a_n} x_0^{a_0} \dots x_n^{a_n}$$

f is called homogeneous of degree d if for any $a_0, \dots, a_n$:

$$\sum a_i = d$$

Example. $f(x, y) = x + y$ is homogenous of degree 1. $f(x, y) = xy + y^2$ is homogenous of degree 2. Clearly, if f is homogenous of degree d , then $f(\lambda x_0, \dots, \lambda x_n) = \lambda^d f(x_0, \dots, x_n)$.

Let f be homogeneous of degree d , let $[x_0 : \dots : x_n] \in \mathbb{P}^n$. Then it makes sense ask whether $f([x_0 : \dots : x_n]) = 0$ or not.

Definition 2.3. A subset $V \subseteq \mathbb{P}^n$ is called a projective variety f if

$$V = \{[x_0 : \dots : x_n] \in \mathbb{P}^n \text{ such that } f_1([x_0 : \dots : x_n]) = \dots = f_m([x_0 : \dots : x_n]) = 0\}$$

for some collection $\{f_1, \dots, f_m\}$ of homogenous polynomials. This is denoted by $V(f_1, \dots, f_m)$.

Remark. $V(x_0, \dots, x_n) = \emptyset$ because $[0 : \dots : 0]$ does not exist in $\mathbb{P}^n$.

Example. Let $V = V(y - x^2) \subseteq \mathbb{A}^2$ affine variety.

$$\begin{aligned}\mathbb{A}^2 &\hookrightarrow \mathbb{P}^2 \\ (x, y) &\mapsto [1 : x : y]\end{aligned}$$

Under this map V becomes $\{[1 : x : y] \in \mathbb{P}^2, y - x^2 = 0\}$. Let $f = zy - x^2 \in k[x, y, z]$ which is homogenous of degree

2. Hence

$V^* = V(f) = \{[x : y : z] \in \mathbb{P}^2 \text{ such that } zy - x^2 = 0\}$ is a projective variety. $V^* \cap \mathbb{A}^2 = V$ but $V^* \neq V$. $\mathbb{P}^2 \setminus \mathbb{A}^2 = \{[0 : x : y]\}$. Notice $[0 : 1 : 0] \in V^* \setminus V$ so $V^* = V \cup \{[0 : 1 : 0]\}$.

Homogenisation: let $f = y - x^2$ where $\deg(f) = 2$. Homogenisation of f : $f^* = zy - x^2 \in k[x, y, z]$ and $f^*(x, y, 1) = f$. In general:

$$f = \sum_{(a_1, \dots, a_n)} \alpha_{a_1, \dots, a_n} x_1^{a_1} \cdots x_n^{a_n} \in k[x_1, \dots, x_n].$$

of degree $d = \max\{\sum a_i\}$.

$$f^* = \sum_{(a_1, \dots, a_n)} \alpha_{a_1, \dots, a_n} x_0^{d - \sum a_i} x_1^{a_1} \cdots x_n^{a_n} \in k[x_1, \dots, x_n].$$

is a homogeneous polynomial such that $f^*(1, x_1, \dots, x_n) = f$. Let $I \subset k[x_1, \dots, x_n]$ be an ideal. Let $I^* = \{f^* \in k[x_0, \dots, x_n], f \in I\}$ is a subset of $k[x_1, \dots, x_n]$.

$$V(I^*) = \{[x_0 : \dots : x_n] \in \mathbb{P}^n, f^*([x_0 : \dots : x_n]) = 0, \forall f^* \in I^*\}$$

is a projective variety.

Definition 2.4. $V(I^*)$ is called the projective closure of $V(I)$. $V(I^*) \setminus V(I)$ are called points at infinity of $V(I)$.

Remark. In general, let $L = \sum_{i=0}^n \alpha_i x_i$ linear function. Let

$$U_L = \{[a_0 : \dots : a_n] : L(a_0, \dots, a_n) \neq 0\}$$

$$U_L \xrightarrow{\text{bijection}} \mathbb{A}^n$$

$$[a_0 : \dots : a_n] \rightarrow \left(\frac{a_1}{L(a_0, \dots, a_n)}, \dots, \frac{a_n}{L(a_0, \dots, a_n)} \right)$$

This identifies $\mathbb{A}^n$ with a subset of $\mathbb{P}^n$.

Let f be a homogenous polynomial in $k[x_0, \dots, x_n]$. Then condition $f[a_0 : \dots : a_n] = 0$ is well-defined. If $f_1, \dots, f_m$ are homogenous polynomials:

$$V(f_1, \dots, f_m) = \{[a_0 : \dots : a_n] \in \mathbb{P}^n : f_i([a_0 : \dots : a_n]) = 0, \forall i\}$$

is called the projective algebraic set. If $I \subset k[x_0, \dots, x_n]$ ideal. I^* is the set of all $f^*, f \in I$. For $f \in k[x_1, \dots, x_n]$ of degree d .

$$f^* = x_0^d f\left(\frac{x_1}{x_0}, \dots, \frac{x_n}{x_0}\right)$$

homogenous of degree d in $k[x_0, \dots, x_n]$.

Example. Let $V(y^2 - x) \subseteq \mathbb{A}^2$. Then

$$f^* = z^2 f\left(\frac{x}{z}, \frac{y}{z}\right) = z^2 \left(\frac{y^2}{z^2} - \frac{y}{z}\right) = y^2 - xz.$$

$V(f^*) = V(y^2 - xz) \subseteq \mathbb{P}^2$ projective variety. $V(f^*) \cap \mathbb{A}^2 = V(f)$. All points $[a : b : c] \in V(f^*)$ where $a \neq 0$. $V(f^*) = V(f) \cup \{[0 : 0 : 1]\}$.

Example. For $f = xy - 1$, $f^* = xy - z^2$, $V(xy - z^2) = V(f^*) \cong V(y^2 - xz)$.

Let $I \subseteq k[x_0, \dots, x_n]$, $V(I)$ is not defined in general in $\mathbb{P}^n$. But if $I = (f_1, \dots, f_n)$ where f_i s are homogeneous, then $V(I) \subseteq \mathbb{P}^n$ is well-defined. An ideal generated by homogeneous polynomials is called a homogeneous ideal.

Let f be a polynomial of degree d , $f = f_d + f_{d-1} + \dots + f_0$ where f_i s are homogeneous of degree i . They are called homogeneous components of f .

Lemma 2.5. An ideal I is homogeneous if and only if for any $f = f_d + f_{d-1} + \dots + f_0 \in I$, $f_i \in I_i$ for any i .

The Lemma implies that any homogeneous ideal is generated by finitely many homogeneous polynomials. By Hilbert Basis Theorem, I is finitely generated, $I = (f_1, \dots, f_r)$. By Lemma, I is generated by the homogeneous components of f_i s. Any $V(I)$ is in fact of the form $V(f_1, \dots, f_r)$ with f_i homogeneous. In particular:

1. If $I \subseteq J$, $V(J) \subseteq V(I)$.
2. $V(I \cap J) = V(I) \cup V(J)$.
3. $[0 : \dots : 0]$ does not exist which means if $I \supset (x_0^N, \dots, x_n^N)$ for some $N \geq 1$ then $V(I) = \emptyset$ in $\mathbb{P}^n$. The converse is not true. For example, $k = \mathbb{R}$, $I = (x^2 + y^2)$.

Zariski topology on $\mathbb{P}^n$. Closed sets: $V(I)$ with I homogeneous. Open sets: their complements. We have an open set

$$D(f) = \{[a_0 : \dots : a_n], f(a_0, \dots, a_n) \neq 0\}$$

In particular, we have $D(x_i)$, $i = 1, \dots, n$. Open sets, we have identified them with $\mathbb{A}^n$ where $\mathbb{A}^n$ viewed as subset of $\mathbb{P}^n$ is open.

Definition 2.6. Quasi-projective variety is open subset of a projective variety i.e. $V(I) \cap U$ where $U \subseteq \mathbb{P}^n$ open.

Example. $\mathbb{A}^n \subseteq \mathbb{P}^n$ is a quasi-projective variety. Similarly,

$$\{[1 : x : y] : y^2 = x\}$$

the affine variety $V(y^2 - x)$ is quasi-projective since it is $V(y^2 - xz) \cap D(z)$.

Recall affine cases

$$\mathbb{A}^n \xrightarrow{\varphi} \mathbb{A}^m$$

$$(a_1, \dots, a_n) \mapsto (\varphi_1(a_1, \dots, a_n), \dots, \varphi_m(a_1, \dots, a_n))$$

where $\varphi_i \in k[x_1, \dots, x_n]$. Let's try to generalise this to a map

$$\mathbb{P}^n \xrightarrow{\varphi} \mathbb{P}^m$$

$$[a_0 : \dots : a_n] \mapsto [\varphi_1(a_0, \dots, a_n) : \dots : \varphi_m(a_0, \dots, a_n)]$$

We must at least assume φ_i s to be homogeneous of same degree. $\varphi([a : b]) = [\varphi_1(a, b) : \varphi_2(a, b)]$. We ll defined means that

$$\varphi([\lambda a : \lambda b]) = \varphi([a : b]) = [\varphi_1(\lambda a, \lambda b) : \varphi_2(\lambda a, \lambda b)] = [\lambda^{\deg \varphi_1} \varphi_1(a, b) : \lambda^{\deg \varphi_2} \varphi_2(a, b)] = [\varphi_1(a, b) : \varphi_2(a, b)].$$

We need another condition:

$$\varphi_0(a_0, \dots, a_n) = \dots = \varphi_m(a_0, \dots, a_n) = 0 \implies a_0 = \dots = a_n = 0.$$

Definition 2.7.

$$\mathbb{P}^n \xrightarrow{\varphi} \mathbb{P}^m$$

$$[a_0 : \dots : a_n] \mapsto [\varphi_0(a_0, \dots, a_n) : \dots : \varphi_m(a_0, \dots, a_n)]$$

is called a polynomial map if φ_i s are homogeneous of same degree. φ_i s are injective for any i .

Remark. For the second condition, it is sufficient that $I = (\varphi_0, \dots, \varphi_n)$ is irrelevant.

Example. Let $k = \mathbb{R}$,

$$\mathbb{P}^2 \rightarrow \mathbb{P}^1$$

$$[z : x : y] \mapsto [z^2 + x^2 : x^2 + y^2]$$

It's well-defined by definition. But if $k = \mathbb{C}$, it's not well-defined.

If $V \subseteq \mathbb{P}^n$, $W \subseteq \mathbb{P}^m$ a polynomial map $V \xrightarrow{\varphi} W$ is $[a_0 : \dots : a_n] \mapsto [\varphi_0, \dots, \varphi_m]$ where φ_i s are homogeneous of same degree and $\varphi_i(a_0, \dots, a_n) = 0$ implies $(a_0, \dots, a_n) = 0$. but you only need to check this for $[a_0 : \dots : a_n] \in V$. In the case of polynomial maps in the affine case, $\varphi : V \rightarrow W$ automatically extends to map $\mathbb{A}^n \rightarrow \mathbb{A}^m$. It is not true in the projective case. It is not true in the projective case: let $\mathbb{P}^1 \subseteq \mathbb{P}^2$ be a closed subset

$$\mathbb{P}^1 \rightarrow \mathbb{P}^1$$

$$[0 : x : y] \mapsto [x : y]$$

which is a well-defined polynomial map. But it doesn't extend to $\mathbb{P}^2 \rightarrow \mathbb{P}^1$ because $[z : x : y] \mapsto [x : y]$ is not defined at point $[1 : 0 : 0]$.

Example.

$$\mathbb{P}^1 \rightarrow \mathbb{P}^2$$

$$[x, y] \mapsto [x^2 : xy : y^2]$$

is a well-defined polynomial map.

Example.

$$\mathbb{P}^1 \xrightarrow{\varphi} \mathbb{P}^2$$

$$[x, y] \mapsto [yx^2 : x^3 : y^3]$$

The image of φ is $V(y^2z - x^3)$.

Remark. If $I \subseteq k[x_1, \dots, x_n]$ is an ideal. It defines $V(I) \subseteq \mathbb{A}^n \subseteq \mathbb{P}^n$. Let $\bar{I}$ be the homogeneous ideal generated by $I^* = \{f^* : f \in I\}$. $V(\bar{I}) \subseteq \mathbb{P}^n$ is called projective closure.

Definition 2.8. φ is an isomorphism if there exists $\psi : W \rightarrow V$ polynomial such that $\psi \circ \varphi = \text{id}_V$, $\varphi \circ \psi = \text{id}_W$.

Example. Take

$$\mathbb{A}^1 \xrightarrow{\varphi} \mathbb{A}^1$$

$$x \mapsto \frac{1}{x}$$

rational map. Then

$$\begin{aligned}\mathbb{P}^1 &\rightarrow \mathbb{P}^1 \\ [1 : x] &\mapsto [1 : \frac{1}{x}] = [x : 1]\end{aligned}$$

The map

$$\begin{aligned}\varphi : \mathbb{P}^1 &\rightarrow \mathbb{P}^1 \\ [y : x] &\mapsto [x : y]\end{aligned}$$

is a polynomial map. Restricted to $\mathbb{A}^1 = \{[1 : x]\}$ it is $x \mapsto \frac{1}{x}$.

Definition 2.9 (Equivalence relation). Let V, W be quasi-projective (Zariski open in projective variety). Let $\varphi : V \rightarrow W$, $\varphi = [\varphi_0 : \dots : \varphi_m]$ such that $\exists x \in V$ where $(\varphi_0(x), \dots, \varphi_m(x)) \neq 0$. We say that $\varphi \sim \psi = [\psi_0 : \dots : \psi_m]$ if

$$\frac{\varphi_i}{\varphi_j} = \frac{\psi_i}{\psi_j}, \quad \forall (i, j) \Leftrightarrow \varphi_i \psi_j = \varphi_j \psi_i.$$

Example. For $m = 1$, $[\varphi_0 : \varphi_1] \sim [\psi_0 : \psi_1]$ if

$$\frac{\varphi_0}{\varphi_1} = \frac{\psi_0}{\psi_1}.$$

If x is such that $[\varphi_0(x) : \dots : \varphi_m(x)]$ and $[\psi_0(x) : \dots : \psi_m(x)]$ exist. Then there exists $\lambda \neq 0$ such that $\varphi_i(x) = \lambda \psi_i(x)$ for all i .

Example. Let

$$\begin{aligned}\mathbb{P}^1 &\xrightarrow{\varphi} \mathbb{P}^1 \\ [x : y] &\mapsto [x : x]\end{aligned}$$

and

$$\begin{aligned}\mathbb{P}^1 &\xrightarrow{\psi} \mathbb{P}^1 \\ [x : y] &\mapsto [1 : 1]\end{aligned}$$

then $\varphi \sim \psi$ since $\frac{x}{x} = \frac{1}{1}$.

Definition 2.10 (Rational Map). Let V, W be quasi-projective variety. Let $\varphi : V \rightarrow W$ is an equivalence class which is represented by

$$[\varphi_0 : \dots : \varphi_m]$$

such that there exists $x \in V$ where $\varphi_i(x)$ are not all zero. Then φ is a rational map.

Example. Any polynomial map is a rational map.

Definition 2.11. Let $\varphi : V \rightarrow W$ be a rational map. Let $x \in V$, x is said to be regular point if φ has a representative $[\varphi_0 : \dots : \varphi_m]$ where not all $\varphi_i(x)$ are zero.

Definition 2.12. Given φ rational map. $\{x : \varphi \text{ is defined}\}$ is called domain of definition of φ .

Proposition 2.13. For φ a rational map, it's domain of definition is Zariski open.

Proof. Let I be the ideal generated by all components of all representatives of φ . I is a homogeneous ideal, $V(I) = \{x : \varphi \text{ is not regular}\}$. Domain of definition of φ is the complement of $V(I)$, it's open. $\square$

Example. Let

$$\begin{aligned}V(yz - x^2) &\xrightarrow{\varphi} \mathbb{P}^1 \\ [z : x : y] &\mapsto [z : x]\end{aligned}$$

φ represents a rational map. This expression is defined by $(z, x) \neq (0, 0)$ but not defined at $[0 : 0 : 1]$. Is φ regular at $[0 : 0 : 1]$? So on $V(yz - x^2)$, we have

$$\varphi = [z : x] \sim [yz, yx] \sim [x^2 : yx] \sim [x : y]$$

Hence, φ is regular at $[0 : 0 : 1]$.

Example. Let

$$\begin{aligned}\mathbb{P}^2 &\xrightarrow{\varphi} \mathbb{P}^1 \\ [z : x : y] &\mapsto [z : x]\end{aligned}$$

Is this map regular at $[0 : 0 : 1]$? Suppose

$$\frac{z}{x} = \frac{\psi_0}{\psi_1} \implies z\psi_1 = x\psi_0 \text{ in } k[x, y, z]$$

Hence, $z \mid \psi_0$, $\psi_0 = z\psi'_0$ which implies $\psi_0(0, 0, 1) = 0$. And $x \mid \psi_1$ implies $\psi_1(0, 0, 1) = 0$, a contradiction. Hence ψ is not regular at $[0 : 0 : 1]$. Domain of distribution of φ is $\mathbb{P}^2 \setminus \{[0 : 0 : 1]\}$.

Remark. The functions in the two examples above are different.

Example.

$$\mathbb{P}^2 \supseteq V(y^2z - x^3) \xrightarrow{\varphi} \mathbb{P}^1$$

$$[z : x : y] \mapsto [x : y]$$

is a rational function defined for $(x, y) \neq (0, 0)$. Is it regular at $[1 : 0 : 0]$? Suppose $\varphi \sim \psi = [\psi_0 : \psi_1]$. Then

$$[x : y] \sim [\psi_0 : \psi_1] \implies x\psi_1 = y\psi_0 \text{ on } V$$

Make $z = 1$ then $x\psi_1(1, x, y) - y\psi_0(1, x, y) = 0$ for all $[1 : x : y] \in V$. But $\{[1 : x : y] \in V\} = V \cap \mathbb{A}^2 = V(y^2 - x^3)$. The polynomial $x\psi_1(1, x, y) - y\psi_0(1, x, y)$ is zero on $V(y^2 - x^3)$. Then

$$x\psi_1(1, x, y) - y\psi_0(1, x, y) = (y^2 - x^3)f$$

for some $f \in k[x, y]$. That is $x(\psi_1(1, x, y) + x^2f) = y(\psi_0(1, x, y) + yf)$. Hence $x \mid \psi_0(1, x, y) + yf$ which implies $\psi_0(1, 0, 0) = 0$ and similarly we have $\psi_1(1, 0, 0) = 0$.

Definition 2.14. $\varphi : V \rightarrow W$, V, W quasi-projective is an isomorphism if

1. φ is regular on V .
2. There is a regular map $\psi : W \rightarrow V$ such that $\psi \circ \varphi = \text{id}_V$ and $\varphi \circ \psi = \text{id}_W$.

Remark. An isomorphism is a bijective map.

Example.

$$\mathbb{P}^1 \xrightarrow{\varphi} \mathbb{P}^1$$

$$[x : y] \mapsto [x^2 : y^2]$$

φ is not an isomorphism because φ is not injective.

Example.

$$\mathbb{P}^1 \xrightarrow{\varphi} \mathbb{P}^1$$

$$[x : y] \mapsto [y : x]$$

Zariski open sets, identified with $\mathbb{A}^n$ by

$$U_i = \{[x_0 : \cdots : x_n] \in \mathbb{P}^n : x_i \neq 0\}$$

$$[x_0 : \cdots : x_n] \mapsto \left(\frac{x_0}{x_i}, \dots, \frac{x_n}{x_i} \right)$$

Then the map $U_i \rightarrow U_j$ is an isomorphism.

Example.

$$V(yz - x^2) \xrightarrow{\varphi} \mathbb{P}^1$$

$$[z : x : y] \mapsto [z : x]$$

Let

$$\Psi : \mathbb{P}^1 \rightarrow V(yz - x^2)$$

$$[s : t] \mapsto [s^2 : st : t^2]$$

Then $\varphi \circ \psi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is the id.

Example. Consider $V = V(y^2z - x^3) \subseteq \mathbb{P}^2$.

$$\mathbb{P}^1 \xrightarrow{\psi} V(y^2z - x^3)$$

$$[s : t] \mapsto [s^3 : st^2 : t^3].$$

$$V(y^2z - x^3) \xrightarrow{\varphi} \mathbb{P}^1$$

$$[z : x : y] \mapsto [x : y]$$

ψ is a rational map which has a rational inverse which is φ . But ψ is not regular at $[0 : 0 : 1]$. But

$$\mathbb{P}^1 \setminus \{[0 : 1]\} \xrightarrow{\psi} V \setminus \{[0 : 0 : 1]\} \xrightarrow{\varphi} \mathbb{P}^1 \setminus \{[0 : 1]\}.$$

Definition 2.15. Let $V \subseteq \mathbb{P}^n, W \subseteq \mathbb{P}^m$ be projective varieties. Let $\varphi : V \rightarrow W$ be a rational map. φ is birational if there exists $U_1 \subseteq V, U_2 \subseteq W$ Zariski open such that $\varphi|_{U_1} : U_1 \rightarrow U_2$ is an isomorphism.

Example. $V(y^2z - x^3) \xrightarrow{\varphi} \mathbb{P}^1$. $U_1 = V(y^2z - x^3) \setminus \{[0 : 0 : 1]\}$ and $U_2 = \mathbb{P}^1 \setminus \{[0 : 1]\}$, φ is an isomorphism $U_1 \rightarrow U_2$ then φ is birational.

Definition 2.16. $\varphi : V \rightarrow W$ is dominant if $\text{Im}(\varphi)$ is Zariski dense in W .

Definition 2.17. $\varphi : V \rightarrow W$ is birational if φ is dominant and φ has a rational inverse.

Exercise. Any Zariski open set is dense.

Exercise. Let φ be a rational map $V \rightarrow W$ regular on U , (open in V). $\varphi|_U$ continuous for Zariski topology.

Exercise. Show that the two definitions of birational are equivalent.

Let $V \subseteq \mathbb{P}^n$ projective variety. Let $f, g \in k[x_0, \dots, x_n]$ homogeneous of same degree then $\frac{f}{g}$ defines a function on $\mathbb{P}^n$:

$$\frac{f(\lambda a_0, \dots, \lambda a_n)}{g(\lambda a_0, \dots, \lambda a_n)} = \frac{f(a_0, \dots, a_n)}{g(a_0, \dots, a_n)}.$$

i.e. $\frac{f([a_0 : \dots : a_n])}{g([a_0 : \dots : a_n])}$ is well-defined. $\frac{f}{g}$ defines a function on $\mathbb{P}^n$ which can be restricted to V provided $g \not\equiv 0$ on V .

Definition 2.18.

$$k(V) = \left\{ \frac{f}{g} \Big|_V : f, g \text{ homogeneous at same degree, } g \text{ is not identically zero on } V \right\} / \sim$$

where $\sim : \frac{f}{g} \sim \frac{f_1}{g_1}$ iff $f g_1 - f_1 g \equiv 0$ on V . Then $k(V)$ defines a field.

Remark. Let $\frac{f}{g} \in k(V)$ such that $f \equiv 0$ on V . We want $\frac{f}{g} \sim \frac{0}{1}$ which follows from definition of $\sim$.

Remark. We have implicitly assumed that V is irreducible. This implies $f g = 0$ on V implies $f = 0$ or $g = 0$ on V .

Example. We have $k(\mathbb{P}^n) \cong k(\mathbb{A}^n)$ via

$$\frac{f(x_0, \dots, x_n)}{g(x_0, \dots, x_n)} \mapsto \frac{f(1, x_1, \dots, x_n)}{g(1, x_1, \dots, x_n)}.$$

Example.

$$\frac{t}{t^2} \sim \frac{\frac{x}{y}}{\frac{x^2}{y^2}} = \frac{y^2(\frac{x}{y})}{y^2(\frac{x^2}{y^2})} = \frac{y}{x^2} \in k(\mathbb{P}^1)$$

More generally, $V \subset \mathbb{P}^n$ projective variety. $V \not\subseteq V(x_0) = \{x_0 = 0\}$, $V_* = V \cap \mathbb{A}^n$ where $\mathbb{A}^n = \{[1 : x_1, \dots, x_n]\}$. Then

$$k(V) \rightarrow k(V_*)$$

$$\frac{f}{g} \mapsto \frac{f(1, x_1, \dots, x_n)}{g(1, x_1, \dots, x_n)}$$

is an isomorphism.

Remark. We have a similar fact for φ^* comparing to the affine case.

Exercise. φ is birational if and only if φ^* is an isomorphism of fields.

The following topics are not examinable: For affine case,

$$\mathbb{A}^n \times \mathbb{A}^m = \mathbb{A}^{n+m}$$

What is $\mathbb{P}^1 \times \mathbb{P}^1$? Then

$$\{[x : y : z : w]\} = \mathbb{P}^3.$$

this is a bad definition.

Definition 2.19 (Segre Embedding).

$$\mathbb{P}^1 \times \mathbb{P}^1 \xrightarrow{\varphi} \mathbb{P}^3$$

$$([x, y], [z, w]) \mapsto [xz : xw : yz : yw]$$

is well-defined.

Indeed,

$$\varphi([\lambda x, \lambda y], [\mu z, \mu w]) = [\lambda \mu x z : \lambda \mu x w : \lambda \mu y z : \lambda \mu y w] = [x z : x w : y z : y w].$$

The image of φ is $V(x_0x_3 - x_1x_2)$ if $\mathbb{P}^3 = \{[x_0 : x_1 : x_2 : x_3]\}$. When doesn't $f([x, y], [z, w]) = 0$ make sense? Bi-homogeneous polynomial $f \in k[x, y, z, w]$. For all $\lambda, \mu \in k$,

$$f(\lambda x, \lambda y, \mu z, \mu w) = \lambda^{d_1} \mu^{d_2} f(x, y, z, w).$$

For f bi-homogeneous, we can define $V(f) \subset \mathbb{P}^1 \times \mathbb{P}^1$ where

$$V(f) = \{([x : y], [z : w]) \in \mathbb{P}^1 \times \mathbb{P}^1 : f([x : y], [z : w]) = 0\}$$

| Example. $f = xz + wy$ is bi-homogeneous of degree $(1, 1)$.

| Example. $f = xz^2 + yw^2$ is bi-homogeneous of degree $(1, 2)$.

Important property of $\mathbb{P}^1 \times \mathbb{P}^1$: it admits 2 maps: $\mathbb{P}^1 \times \mathbb{P}^1 \xrightarrow{\pi_i} \mathbb{P}^1$ which are projections.

Exercise. There is no non-constant polynomial map from $\mathbb{P}^2 \rightarrow \mathbb{P}^1$.

Generally,

$$\mathbb{P}^n \times \mathbb{P}^m \rightarrow \mathbb{P}^{(n+1)(m+1)-1}([x_0 : \dots : x_n], [y_0 : \dots : y_m]) \mapsto [x_0y_0 : \dots : x_0y_m : x_1y_0 : \dots : x_1y_m : \dots : x_ny_0 : \dots : x_ny_m]$$

also known as Segre Embedding. And

$$\text{Im } \varphi = V(z_{ij}z_{kl} - z_{il}z_{kj} : 0 \leq i, k \leq m, 0 \leq j, l \leq n)$$

Blow-up: Let

$$\mathbb{A}^2 \times \mathbb{P}^1 = \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1\}$$

An algebraic subset of $\mathbb{A}^2 \times \mathbb{P}^1$ is defined as zeros of polynomial: $f(x, y, \lambda s, \lambda t) = \lambda^d f(x, y, s, t)$. Blow up of $\mathbb{A}^2$ at $(0, 0)$ is the subset of $\mathbb{A}^2 \times \mathbb{P}^1$ defined as

$$\tilde{\mathbb{A}}^2 = \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 \text{ such that } xt - ys = 0.\}$$

If $(x, y) \neq (0, 0)$, then $(x, y) \in \text{line } [s : t]$. Therefore,

$$\tilde{\mathbb{A}}^2 = \{(P, \ell) \in \mathbb{A}^2 \times \mathbb{P}^1 \text{ lines through } P : P \in \ell\}$$

Consider $\tilde{\mathbb{A}}^2 \xrightarrow{\pi} \mathbb{A}^2$. Then

$$\pi^{-1}(0) = \{(0, 0), [s : t]\} = \mathbb{P}^1$$

known as the "Exceptional divisor".